

CSE227:ADVANCED ANDROID APP DEVELOPMENT

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: illustrate the use of Realtime Database for dynamic and responsive user experiences.

CO2 :: develop interactive visuals with Canvas to enhance user engagement.

CO3 :: demonstrate retrofit for handling complex networking tasks in modern applications.

CO4 :: design an AI-based chat interface for user–AI interaction.

CO5 :: analyze device connectivity using Bluetooth, Wi-Fi, and various sensors in real-time applications.

CO6 :: design and manage web content using web apps.

Unit I

Building Cloud-Enabled Applications with Firebase : Introduction of Firebase, Feature of Firebase, connecting firebase to app, Firebase Authentication, Firebase UI, Real time dataset base configure and setup, CRUD operation, Firebase Notification

Unit II

Advanced Graphics : Canvas, Animations, Transitions

Unit III

Advanced Networking with Retrofit : Retrofit Basics and Integration, Handling API Requests, Parsing and Displaying Data, Asynchronous Networking, Authentication

Unit IV

AI Integration in Android Applications : Introduction to AI in mobile apps, Role of AI assistants in Android, Overview of ChatGPT OpenAI and Gemini Google, Basics of AI APIs REST APIs and API keys, Integrating Google Gemini API in Android, Generating AI text responses, Handling user input and AI output, Basic error handling and API usage limits

Unit V

Smart Device Communication and Sensing Systems : Bluetooth overview, Perform network operations, Available Devices, Paired Devices Wi-Fi Companion device pairing overview, Connect devices, Motion sensors, Position sensors, Environment sensors

Unit VI

Web-Based Content : Building and Managing Web Apps in WebView, Migrating to WebView in Android, Supporting Different Screens in web Apps, Advance android programming

List of Practicals / Experiments:

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- Firebase: Create an application based on Firebase Database
- Advanced Graphics: Create an application based on canvas, use the draw methods to draw circle, rectangle etc. and apply some animations.
- Sensors: Create an application based on different types of sensors.
- Connectivity: Create an application which will turn on and off your Bluetooth device and show nearby Bluetooth devices.
- Web-Based Content: Create an application to show the content in the WebView.
- Networking with Retrofit & Ktor: Create an application based on Networking with Retrofit & Ktor .

Text Books:

1. . THE ANDROID DEVELOPER'S COOKBOOK: BUILDING APPLICATIONS WITH THE ANDROID SDK by RONAN SCHWARZ, PEARSON

References:

1. ANDROID APPLICATION DEVELOPMENT ALL-IN-ONE FOR DUMMIES by BARRY BURD, JOHN PAUL MUELLER, WILEY

